A centralizer bound for symplectic strings in dimension four

Human Compression project

We prove an upper bound for connected strings of involutions in characteristic two by determining two mutual centralizers. The proof uses Jordan normal form and elementary matrix calculations.

Lemma 1. *Over an algebraically closed field k of characteristic two, two noncommuting involutions in $\mathrm{SL}_2(k)$ have no common invariant line, and their common centralizer in $M_2(k)$ consists of scalars. The centralizer in $\mathrm{SL}_2(k)$ of a single nonidentity involution is abelian.*

Proof. A nonidentity involution is unipotent with one fixed line. If two involutions preserved the same line, both would be upper unipotent in a suitable basis and would commute. Otherwise use their distinct fixed lines as a basis. Their matrices are

$$u = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 & 0 \\ \beta & 1 \end{pmatrix}, \qquad \alpha\beta \neq 0.$$

Commuting with u forces a matrix to have the form $\left(\begin{smallmatrix} c & d \\ 0 & c \end{smallmatrix}\right)$; commuting also with v forces $d = 0$. In determinant one the first centralizer has $c^2 = 1$, hence $c = 1$, and is abelian. □

Lemma 2. *Let V be a four-dimensional symplectic space over k. Suppose $A = \langle a, b\rangle$ and $D = \langle e, f\rangle$ are generated by noncommuting involutions and commute elementwise with each other. Then the full centralizers $C_{\mathrm{Sp}(V)}(A)$ and $C_{\mathrm{Sp}(V)}(D)$ commute elementwise.*

Proof. Put $h = ab$. Since a, b do not commute, $h^2 \neq 1$. The symplectic form pairs generalized eigenspaces only when their eigenvalues are reciprocal. Thus the multiplicities of λ and λ^{-1} agree, and the eigenvalue-one primary space is nondegenerate. Also $aha = h^{-1}$.

First, if h is not semisimple, its symplectic centralizer is abelian. Here are all possibilities, including their justification in dimension four. If all eigenvalues are one, a Jordan block has size at least three, since otherwise $h^2 = 1$. The partition $3 + 1$ is impossible. Indeed write $hx_1 = x_1$, $hx_2 = x_2 + x_1$, $hx_3 = x_3 + x_2$ and $hy = y$. Preserving the pairings of (x_1, x_3), (x_2, x_3) and (x_2, y) successively gives $B(x_1, x_2) = B(x_1, x_3) = B(x_1, y) = 0$. Together with alternation this makes x_1 radical. The only remaining case is one Jordan block of size four, whose matrix centralizer consists of polynomials in that block and is abelian.

If a nonunit eigenvalue has a nontrivial Jordan block, reciprocal pairing uses all four dimensions: there are two size-two blocks at λ, λ^{-1}, $\lambda \neq 1$. A commuting symplectic map is determined by its action on the first block; that action centralizes a size-two Jordan block and lies in its abelian polynomial algebra. The other possibility has a size-two unipotent block on the eigenvalue-one plane and simple eigenvalues λ, λ^{-1} on its perpendicular plane. Its symplectic centralizer is a product of the abelian centralizer in Lemma 1 and a one-dimensional torus. These exhaust the nonsemisimple cases by reciprocal multiplicities. Since D is nonabelian and centralizes h, none can occur.

Thus h is semisimple. Its centralizer can be nonabelian only for one of the spectra

$$(\lambda, \lambda^{-1}, 1, 1), \qquad (\lambda, \lambda, \lambda^{-1}, \lambda^{-1}), \qquad \lambda \neq 1. \tag{1}$$

Indeed four simple nonunit eigenvalues give only a torus, and four unit eigenvalues would give $h = 1$.

In the first case write $V = W \perp Z$, where $h|_Z = 1$ and W is its nonunit plane. The involution a preserves these planes and exchanges the two eigenlines in W. A matrix commuting with both $a|_W, h|_W$ is scalar, and a symplectic scalar in characteristic two is the identity. Consequently

$$C_{\mathrm{Sp}(V)}(A) = 1_W \times C_{\mathrm{Sp}(Z)}(a|_Z).$$

If $a|_Z \neq 1$, this group is abelian by Lemma 1, impossible because it contains D. Hence $a|_Z = b|_Z = 1$ and this centralizer is exactly $1_W \times \mathrm{Sp}(Z)$. The two generators of D are noncommuting involutions on Z. Lemma 1 says that they have no common fixed vector and have only scalar commuting matrices on Z. There are therefore no intertwining blocks between their trivial action on W and their action on Z: the absence of a fixed covector follows as well from the invariant symplectic form on Z. It follows that $C_{\mathrm{Sp}(V)}(D) = \mathrm{Sp}(W) \times 1_Z$. The two asserted centralizers commute.

In the second case let E, E' be the two eigenspaces of h. Choose a basis of E and its image under a as basis of E'. Then a exchanges the two coordinate planes and the form matrix is

$$\begin{pmatrix} 0 & S \\ S & 0 \end{pmatrix},$$

where S is invertible and symmetric; symmetry follows from invariance under a and alternation in characteristic two. The centralizer of A consists exactly of $\mathrm{diag}(T, T)$ with $T^{\mathsf{T}} S T = S$. If S is nonalternating, orthogonalization and square roots in k put it into the form I_2. Its isometries are exactly

$$T = \begin{pmatrix} t & 1+t \\ 1+t & t \end{pmatrix} \quad (t \in k),$$

an abelian group. This is again impossible because it contains D.

Thus S is alternating; normalize it to $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $C_{\mathrm{Sp}(V)}(A) = \{\mathrm{diag}(T, T) : T \in \mathrm{SL}_2(k)\}$. The two matrices representing e, f on E are noncommuting involutions. By Lemma 1, every block of an endomorphism commuting with D is scalar. Such an endomorphism is a two-by-two block matrix $(c_{ij} I_2)$. It preserves the displayed form precisely when $(c_{ij})^{\mathsf{T}} J (c_{ij}) = J$. Every such block matrix commutes with every $\mathrm{diag}(T, T)$. This proves the lemma in the final case. □

Theorem 3. *Over any field of characteristic two, a sequence of involutions in a four-dimensional symplectic group satisfying*

$$[\rho_i, \rho_j] = 1 \quad (|i - j| > 1), \qquad [\rho_i, \rho_{i+1}] \neq 1$$

has length at most five.

Proof. Extend scalars to an algebraic closure, which preserves matrix equalities and inequalities. If six generators existed, put $A = \langle \rho_0, \rho_1 \rangle$ and $D = \langle \rho_4, \rho_5 \rangle$. They satisfy Lemma 2. But $\rho_2 \in C(D)$ and $\rho_3 \in C(A)$, so ρ_2 and ρ_3 commute, a contradiction. □

Exact source interface and attribution. For $q = 2^k \geq 4$, every string C-marking of $\mathrm{Sp}(4, q)$ is connected: a disconnected string would generate two nontrivial commuting normal subgroups, impossible in this simple group. Thus Theorem 3 gives the upper bound five in Brooksbank–Ferrara–Leemans, *Orthogonal groups in characteristic 2 acting on polytopes of high rank*, arXiv:1709.02219v2, Theorem 6.1. Simplicity of $\mathrm{Sp}(4, q)$ for $q \geq 4$ is a retained classical input only for this final interface, not either lemma or Theorem 3. The source's existence and small-field assertions are separate obligations. This component proves no existence assertion or small-field classification.